

Midterm Exam – Solution

Microeconomic Theory III – ECON 709, Fall 2023

Do not open until instructed. Closed-book exam.

The exam consists of 2 questions with subquestions each. In total, you can reach 60 points; that is, 30 points for each question. The exam is from 10.05–11.25am. Good luck!

1. (30 points) *Choice* (after Rubinstein, A13)

A decision maker has in mind several criteria $(\succ_i)_{i \in I}$ where each $\succ_i$ is a strict ordering (complete and transitive) of the elements in a finite set X . Whenever the agent chooses from a set $A \subseteq X$, he is satisfied with any element $a \in A$ such that for any other $b \in A$, there is some i (i probably depends on b) for which $a \succ_i b$. That is, the choice correspondence C that is induced by this procedure is defined by $a \in C(A)$ if $a \succ_i b$ for some $i \in I$, for any other $b \in A$.

Thus, for example, if he has only one criterion in mind then the induced choice correspondence picks the unique maximal element from each set; if he has two in mind, where one is the negation of the other ($a \succ_1 b \Leftrightarrow b \succ_2 a$), then the induced choice correspondence is $C(A) \equiv A$.

- a) (5 points) Show that if $a \in C(A) \cap C(B)$, then $a \in C(A \cup B)$.

Take $a \in C(A) \cap C(B)$. Then $a \in C(A)$ and $a \in C(B)$ and hence $a \succ_i b$ for some i for all $b \neq a$ in A and B . Then, $a \succ_i b$ for some i for all $b \neq a$ in $A \cup B$. Thus $a \in C(A \cup B)$.

- b) (4 points) Show that the induced choice correspondence satisfies Sen's α , that is, if $a \in A \subseteq B$ and $a \in C(B)$, then $a \in C(A)$.

Take $a \in C(B)$. Then $a \succ_i b$ for some i for all $b \neq a$ in B . Thus for any $A \subseteq B$, $a \succ_i b$ for some i for all $b \neq a$ in A . Hence, $a \in C(A)$.

- c) (12 points) Show that the induced choice correspondence satisfies Path Independence, that is, $C(A) = C(C(A_1) \cup C(A_2))$, for any $A \in X$ and any $\{A_1, A_2\}$ that is a partition of A .

Take $a \in C(A)$. Then $a \succ_i b$ for some i for all $b \neq a$ in $A = \{A_1, A_2\}$. Thus if $a \in A_1$, then $a \in C(A_1)$; if $a \in A_2$, then $a \in C(A_2)$. Also, $a \in C(C(A_1) \cup C(A_2))$ as $C(A_1) \cup C(A_2) \subseteq A$.

Take $a \in C(C(A_1) \cup C(A_2))$ and wlog. let $a \in A_1$. Then, $a \succ_i b$ for some i for all $b \neq a$ in A_1 and for all $b \neq a$ in $C(A_2)$.

Suppose by contradiction that $a \notin C(A)$. Then there must exist $c_1^* \in A_2 \setminus C(A_2)$ such that $c_1^* \succ_i a$ for all i . Since $c_1^* \in A_2 \setminus C(A_2)$, there must be $c_2^* \in A_2$ such that $c_2^* \succ_i c_1^*$ for all i and thus $c_2^* \succ_i c_1^* \succ_i a$ for all i . If $c_2^* \in C(A_2)$, we have a contradiction to $a \in C(C(A_1) \cup C(A_2))$. If $c_2^* \notin C(A_2)$, there must exist $c_3^* \in A_2 \dots$ and so on. Since $A_2 \setminus C(A_2)$ is finite, at some point c_k^* must be in $C(A_2)$, which leads to our contradiction.

- d) (9 points) Show that there exists a decision maker of the type described above for whom the induced choice correspondence is not rationalizable.

Suppose $I = \{\succ_1, \succ_2\}$ and $X = \{x, y, z\}$. Let $y \succ_1 z \succ_1 x$ and $z \succ_2 x \succ_2 y$. Then $C(\{x, y\}) = \{x, y\}$ and $C\{x, y, z\} = \{y, z\}$ which violates the Weak Axiom.

2. (30 points) *Consumer preferences and Walrasian demand*

Define a preference relation $\succsim$ on $\mathbb{R}_+^3$ by

$$(x_1, x_2, x_3) \succsim (y_1, y_2, y_3) \iff \text{median}\{x_1, x_2, x_3\} \geq \text{median}\{y_1, y_2, y_3\}.$$

That is, x is weakly preferred over y if and only if the intermediate value of x is weakly larger than the intermediate value of y . For example, $(5, 2, 10) \succsim (2, 4, 8)$ because $\text{median}\{5, 2, 10\} = 5 \geq \text{median}\{2, 4, 8\} = 4$. This preference relation is complete, transitive, and monotone.

- a) (4 points) Is this preference relation strongly monotone? If so, prove it; if not give a counter-example.

No. $(3, 4, 5) \sim (3, 4, 6)$.

- b) (6 points) Is this preference relation convex? If so, prove it; if not give a counter-example.

No. E.g. $(3, 9, 14) \succ (3, 13, 6)$ but $1/2 * (3, 13, 6) + 1/2 * (3, 9, 14) = (3, 11, 10) \succ (3, 9, 14)$.

- c) (8 points) Using the definition of continuity of preferences, show that this preference relation is continuous.

Take $x \succ y$. Then $\text{med}(x) > \text{med}(y)$. Take $\epsilon \in (0, (\text{med}(x) - \text{med}(y))/2)$. Consider the ϵ -ball around x , $B_\epsilon(x)$, and the ϵ -ball around y , $B_\epsilon(y)$.

Claim: $\text{med}(x') \in [\text{med}(x) - \epsilon, \text{med}(x) + \epsilon]$ for any $x' \in B_\epsilon(x)$.

Proof: Let $x_k \leq x_l \leq x_m$, and thus $\text{med}(x) = x_l$. Observe that $x'_j \in [x_j - \epsilon, x_j + \epsilon]$ for $j = k, l, m$.

Suppose $\text{med}(x') = x'_l$. Then immediately $\text{med}(x') \in [\text{med}(x) - \epsilon, \text{med}(x) + \epsilon]$.

Suppose second that $med(x') = x'_k$. If $x'_l \leq x'_k \leq x'_m$, then $x_l - \epsilon \leq x'_k \leq x_k + \epsilon \leq x_l + \epsilon$. If $x'_m \leq x'_k \leq x'_l$, then $x_l - \epsilon \leq x_m - \epsilon \leq x'_k \leq x_l + \epsilon$.

Suppose third that $med(x') = x'_m$. If $x'_k \leq x'_m \leq x'_l$, then $x_l - \epsilon \leq x_m - \epsilon \leq x'_m \leq x_l + \epsilon$. If $x'_l \leq x'_m \leq x'_k$, then $x_l - \epsilon \leq x'_m \leq x_k + \epsilon \leq x_l + \epsilon$.

Similarly, $med(y') \in [med(y) - \epsilon, med(y) + \epsilon]$ for any $y' \in B_\epsilon(y)$.

Thus, $med(x') > med(y')$ and $x' \succ y'$.

- d) (12 points) Determine the Walrasian demand correspondence for this preference relation, as a function of prices p_1, p_2, p_3 and wealth w . To make life simpler, you can assume $p_1 \geq p_2 \geq p_3 > 0$.

Claim: $x_1(p, w) = 0, x_2(p, w) = x_3(p, w) = w/(p_2 + p_3)$ if $p_1 > p_2 \geq p_3$.

Proof:

The budget constraint binds at the optimum because preferences are monotone.

Consider x with $x_i \leq x_j < x_k$. Then $med(x) = x_j$. Take x' with $x'_i = x_i, x'_j = x_j + \epsilon p_k/p_j < x'_k = x_k - \epsilon$. Then $x' \in B_{p,w}$ and $med(x') = x'_j > med(x)$. Thus we must have $x_i \leq x_j = x_k$.

Now consider x with $0 < x_i \leq x_j = x_k$. Take x' with $x'_i = 0, x'_j = x_j + x_i p_i/(2p_j)$, and $x'_k = x_k + x_i p_i/(2p_k)$. Then $x' \in B_{p,w}$ and $med(x') > med(x)$. Thus we must have $0 = x_i < x_j = x_k$.

Suppose $x_1 \neq 0$. Then $x_1 = x_j = w/(p_1 + p_j)$ and $x_i = 0$. Now consider x' with $x'_1 = 0$ and $x'_2 = x'_3 = w/(p_2 + p_3)$. Then $x' \in B_{p,w}$ and $med(x') > med(x)$ because $p_2 + p_3 < p_1 + p_j$. This completes the proof.

Note that for $p_1 = p_2 \geq p_3$, any bundle where any two cheapest goods get consumed in equal amounts, and a most expensive good is not consumed is optimal.